Web application such as the login session, the app's response time, etc.
7. Performance of the app when many users hit it simultaneously (load testing).
8. Security features of the app to ensure that the user's data accessed by the app is secure.

Role of open source tools in optimising the performance of various Web applications

There are many considerations that software quality assurance engineers need to take into account and evaluate before giving a thumbs-up to any Web app. Since we are in an era of automation, a machine or system helps us do all our mundane manual activities. There are various open source tools that can be used to evaluate and optimise the performance of any Web application on the basis of the users' requirements (and they let users know if improvements are required). Let's look at how open source tools assist in optimising the performance of Web applications by first evaluating the apps and then removing the bugs discovered in the process.

1. An automation tool helps a tester in generating and executing different test cases and scenarios for the specific Web app under test.
2. A tester can even record various actions that need to be performed while validating any Web app in the form of code or script, and this can be executed while testing that scenario again during regression testing.
3. Developers of Web apps also need to perform unit testing, and this is done using different open source tools like Selenium, Watir, etc.
4. Open source tools like Selenium and AutoIt help in performing functional testing for any Web app by validating the features of different elements in the app.
5. Validation of the GUIs in different tabs or pages of a Web app needs to be done to check if these are visible in the required format, for which open source tools like AutoIt are apt.
6. Once different modules of the app are tested separately, they also need to be tested after integration; for this, integration or link testing for the complete app is done with different open source tools.
7. It's even possible to fetch data from a specific database or data warehouse and perform validation on it using open source automation tools. This ensures that the data fetched by the application is what users need or asked for.
8. Every Web app developed needs to ensure its users are secure. It's quite important to users that their data is secure and not accessible by anyone not authorised to do so. An open source tool like SOAPUI is good enough to evaluate the security parameters of any Web app.
9. Performance and load testing can also be performed using open source tools like JMeter, to ensure that the app does not crash under a heavy load and works as per the user's expectations.
10. Detailed tracking and reporting of the test cases, whether these were passed or failed, as well as information about the bugs detected can also be retrieved using open source tools.

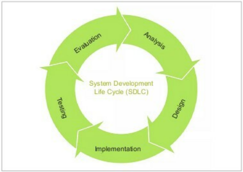

Figure 1: SDLC flow diagram (Image credits: Google images)

Figure 2: Workflow diagram for testing a Web app (Image credits: Google images)

Top five open source tools for evaluating and optimising Web apps

Here are the best open source tools that are widely used to test and evaluate any Web application.

Selenium

Selenium is an open source software testing framework designed for Web apps. It facilitates in authoring tests without even learning a test scripting language (Selenium IDE) by providing a record/playback tool. Selenium also provides a test domain-specific language called Selenese, due to which we can write tests in different popular programming languages like C#, Java, Groovy, Peru, Python, PP, Ruby and Scala. The automated tests can then be run against most of the modern Web browsers. Selenium can be deployed on Windows, Linux and OS X platforms. It is open source software released under the Apache 2.0 licence.

Supported scripting languages: Java, C#, Ruby, Groovy, PEEL, Python, PHP and Scala

Features:

1. Provides a record/playback tool to automate test suites.
2. Identifies components of the app using the properties of various elements.
3. Comprises the Selenium IDE, the client API, Web driver, the grid and remote control.
4. Commands generated in the Selenium IDE Table mode, also known as Selenium commands, are simple type or click-type commands.